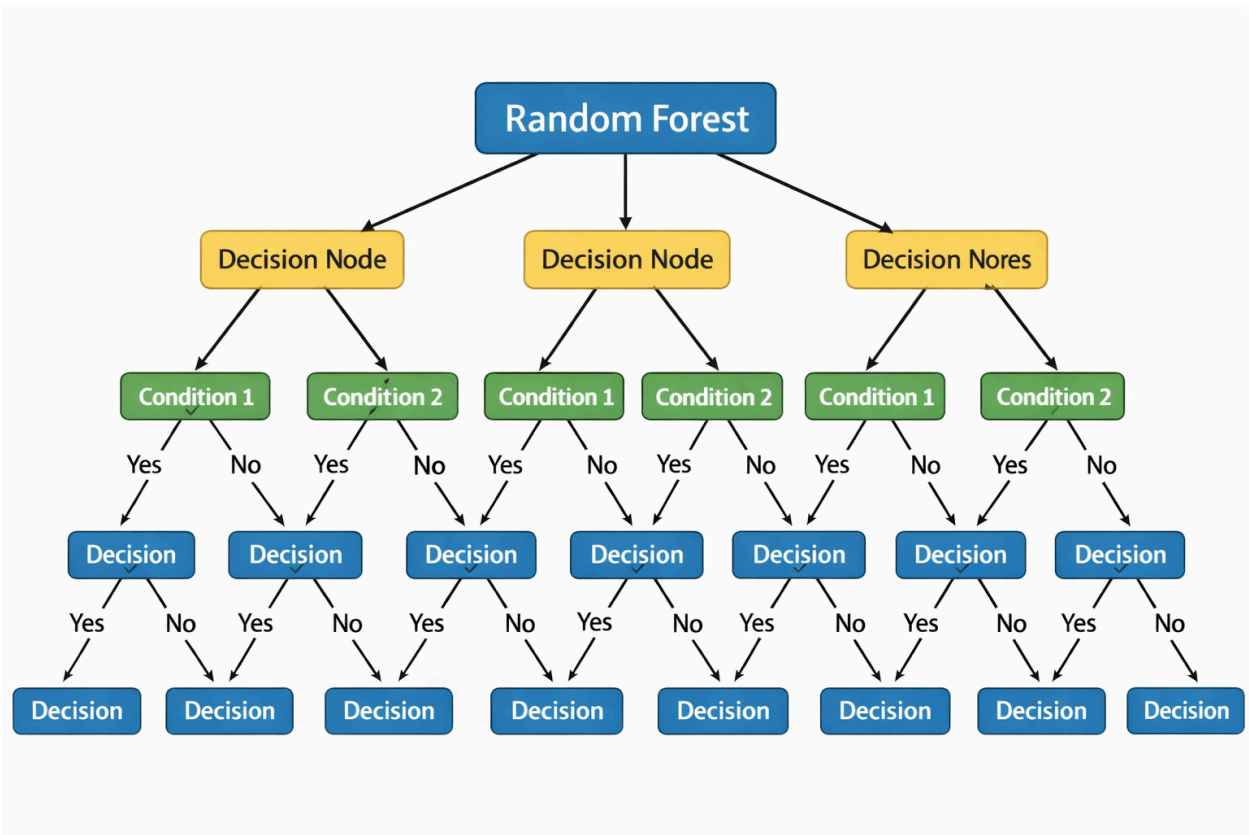
Documentation-Random Forest

Random Forest is a machine learning algorithm that builds multiple decision trees and combines their predictions to produce a more accurate and stable final result.

It works by averaging the outputs (for regression) or taking a majority vote (for classification), which helps reduce overfitting and improves overall performance.



1 Problem Formulation

Input Space:
$X \in R^d$ (numerical features) and/or categorical feature space

Output Space:

- Classification: $Y \in \{1, 2, \ldots, K\}$
- Regression: $Y \in R$

Data Distribution:
Training samples $\{(x_i, y_i)\}$ for $i = 1$ to n are assumed i.i.d. from unknown joint distribution $P(X, Y)$.

Learning Objective (Expected Risk Minimization):

Find a function $f(X)$ that minimizes:

$$\text{Expected Risk} = E[L(Y, f(X))]$$

Since true distribution is unknown, minimize empirical risk via ensemble averaging.

2 Model Specification

Random Forest is an ensemble of B decision trees.

Hypothesis Function:

For regression:

$$f(x) = (1/B) \Sigma T_b(x)$$

For classification:

$$f(x) = \text{majority_vote } \{T_b(x)\}$$

Where:

- $T_b(x)$ = prediction of b-th tree
- B = number of trees

Parameter Space:

- Tree structures for each tree
- Bootstrap samples
- Feature subset size (m_try)
- Number of trees B
- Tree depth and node constraints

Structural Assumptions:

- Trees are grown independently
- Each tree trained on bootstrap sample

- Random subset of features considered at each split
- Non-linear, non-parametric model

3 Loss Function

Same as decision trees but aggregated across ensemble.

For Classification

Gini impurity:

$$\text{Gini} = 1 - \Sigma\, p_k^2$$

Entropy:

$$\text{Entropy} = -\Sigma\, p_k \log p_k$$

For Regression

Mean Squared Error (MSE):

$$\text{MSE} = (1/n)\, \Sigma\, (y_i - y_{mean})^2$$

Why this loss? (Statistical Reasoning)

- Individual trees minimize impurity locally.
- Ensemble averaging reduces variance.
- Majority vote approximates Bayes classifier under large B.

Convexity Properties:

- Individual trees are non-convex.
- Ensemble does not make global objective convex.
- Still a non-convex learning system.

4 Objective Function

Empirical Risk:

$$(1/n)\, \Sigma\, L(y_i, f(x_i))$$

Where:

$f(x)$ = aggregated prediction across trees.

Regularized Formulation:

Random Forest regularizes implicitly via:

- Bootstrap sampling
- Feature randomness
- Averaging predictions

No explicit penalty term, but variance reduction acts as regularization.

5 Optimization Method

Closed-form or Iterative?

Iterative ensemble building:

1. Draw bootstrap sample.
2. Train full decision tree.
3. Repeat B times.
4. Aggregate predictions.

Gradient Expression:

Not gradient-based.

Convergence Guarantees:

As $B \to \infty$:

Variance of ensemble prediction decreases:

$$\mathrm{Var_RF} \approx \rho\, \sigma^2 + (1 - \rho)/B\, \sigma^2$$

Where:

- ρ = correlation between trees
- σ^2 = variance of individual tree

As B increases, second term $\to 0$.

Computational Complexity:

Training complexity:

$O(B\ n\ d \log n)$

Parallelizable across trees.

6 Statistical Interpretation

MLE / MAP Connection:

Each tree approximates conditional distribution $P(Y \mid X)$.
Ensemble approximates expectation over bootstrap distributions.

Random Forest can be viewed as:

Bagging estimator approximating:

$E_{bootstrap}[\ T(X)\ ]$

Noise Model Assumption:

Regression assumes Gaussian noise (via MSE).
Classification approximates multinomial distribution.

Probabilistic Meaning of Outputs:

Classification probability:

$P(Y = k \mid X) \approx$ proportion of trees predicting class k

Regression output approximates:

$E[Y \mid X]$

7 Regularization & Generalization

Bias–Variance Tradeoff:

- Individual tree: Low bias, high variance
- Random Forest: Slightly higher bias, much lower variance

Variance reduction via averaging:

$Var(mean) = Var(tree)/B$ (if independent)

Overfitting Behavior:

- Rarely overfits as B increases
- Deep trees acceptable due to averaging

Capacity Control Mechanism:

- Number of trees (B)
- Maximum depth
- m_try (feature subset size)
- Minimum samples per leaf

8 Theoretical Properties

Convexity / Global Optimality:

- Non-convex model
- No global optimality guarantee

Consistency:

Under certain assumptions:

Random Forest is consistent as:

- $n \to \infty$
- $B \to \infty$

Stability Considerations:

- More stable than single tree
- Reduced sensitivity to small data perturbations

9 Computational Complexity

Training Time Complexity:

$O(B\, n\, d \log n)$

Parallelizable → practical efficiency.

Inference Time Complexity:

$O(B \times tree_depth)$

Memory Complexity:

$O(B \times number_of_nodes_per_tree)$

10 Limitations

When it fails:

- Very high-dimensional sparse data (e.g., text)
- Extrapolation beyond training range (regression)
- Extremely small datasets

Assumption Violations:

- Axis-aligned splits limit representation of oblique boundaries

Sensitivity Issues:

- Biased toward features with many categories
- Large memory usage
- Less interpretable than single tree